

MODULE 3 - PREVIOUS YEAR QUESTIONS WITH ANSWERS

PART A - Short Answer Questions

Q1) Define sample space and event with example.

Answer: Sample Space: Set of all possible outcomes. Example: $S=\{HH,HT,TH,TT\}$. Event: Any subset of S . Example: $A=\{HH\}$.

Q2) State formula for conditional probability.

Answer: $P(A|B)=P(A\cap B)/P(B)$.

Q3) State Bayes theorem.

Answer: $P(A|B)=[P(B|A)P(A)]/P(B)$.

Q4) Define independence.

Answer: Events A and B are independent if $P(A\cap B)=P(A)P(B)$.

Q5) Define PMF and PDF.

Answer: PMF: $P(X=x)=f(x)$, $\sum f(x)=1$. PDF: $f(x)\geq 0$ and $\int f(x)dx=1$.

PART B - Descriptive Problems

Q6) Find mean and variance for $x=0,1,2$ with $P(x)=1/4,2/4,1/4$.

Answer: Mean=1. Variance=0.5.

Q7) A coin is tossed 5 times. Find probability of exactly 3 heads.

Answer: $P(3)=5C3(1/2)^5=5/16$.

Q8) State mean and variance of Binomial distribution.

Answer: Mean=np, Variance=np(1-p).

Q9) If average calls per hour is 3, find $P(X=2)$ using Poisson.

Answer: $P(2)=e^{-3} \cdot 3^2 / 2! \approx 0.224$.

Q10) State mean and variance of Poisson distribution.

Answer: Mean= λ , Variance= λ .

Q11) If $p=0.3$, find probability first success occurs on 4th trial.

Answer: $P(4)=(0.7)^3(0.3)=0.1029$.

Q12) Find mean of Geometric distribution.

Answer: Mean=1/p.

Q13) State PDF of Normal distribution.

Answer: $f(x)=1/\sqrt{2\pi\sigma^2} e^{-\{(x-\mu)^2/(2\sigma^2)\}}$. Mean= μ , Variance= σ^2 .

PART C - Long Answer Questions

Q14) Derive mean of Binomial distribution.

Answer: Let $X = \sum X_i$ where $X_i \sim \text{Bernoulli}(p)$. Then $E(X) = \sum E(X_i) = np$.

Q15) Show that Poisson is limiting form of Binomial.

Answer: If $n \rightarrow \infty$, $p \rightarrow 0$, $np = \lambda$, then Binomial tends to Poisson: $P(X=x) = e^{-\lambda} \lambda^x / x!$.